

Spatiotemporal distribution patterns of large and medium-sized mammals in a biodiversity hotspot: Implications for conservation

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ARTICLE INFO

Keywords:

Camera trap
Gaoligong mountains
Human disturbance
Low-altitude habitats
Occupancy modeling
Temporal beta diversity

ABSTRACT

Understanding wildlife distribution patterns in relation to human disturbance and environmental variables across varying spatial and temporal dimensions is crucial for advancing effective conservation strategies. Here, we employed multi-species occupancy models on an extensive dataset derived from a systematic camera trapping survey across the Gaoligong Mountains to examine the effects of human modification (settlements and farmland) and human presence (camera detections of humans and livestock), latitude, and elevation on the diversity and distribution of large and medium-sized mammal species. We divided the study area into three sections (northern, middle and southern), 12 altitude zones, and sampling years into wet and dry seasons. We used spatial and temporal β diversity indices to quantify changes in species assemblages. From 375 stations over 113,204 camera days, we recorded 34 large and medium-sized mammal species. Throughout the year, the community mean occupancy showed a strong negative response to latitude. The dry season community mean occupancy negatively responded to latitude, elevation, and human modification and presence. Species turnover was the primary driver of diversity changes across all sections. Importantly, lower altitude zones from 1700 to 2300 m gained more species in the dry seasons, compared with wet seasons. These findings underscore the critical importance of low-elevation habitats for mammal conservation, emphasizing the need for targeted protection and restoration efforts in these areas. Effective conservation strategies should incorporate spatial heterogeneity and temporal dynamics, with particular attention to the varying habitat requirements of species across different seasons.

1. Introduction

Human activities have dramatically accelerated the loss of global

biodiversity in the Anthropocene, triggering the onset of the sixth mass extinction crisis (Johnson et al., 2017; Pimm et al., 2014). To mitigate or reverse these trends, comprehensive analyses of species distributions,

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<https://doi.org/10.1016/j.biocon.2024.110863>

Received 29 April 2024; Received in revised form 12 October 2024; Accepted 5 November 2024

Available online 22 November 2024

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population dynamics, community shifts, and responses to anthropogenic pressures are essential (Li et al., 2018; McMahon et al., 2011). Such analyses are critical for devising and implementing effective conservation strategies. Human activities and environmental variables are among the main factors affecting the spatiotemporal patterns of large and medium-sized mammals (Johnson and Hering, 2009; Li et al., 2024b; Li et al., 2022b; Li et al., 2022c; Pillay et al., 2022). Notably, the response of wildlife to human activities depends on the type and intensity of such activities and exhibits interspecies variability (Li et al., 2022c; Nickel et al., 2020; Tucker et al., 2018). Predator-prey pressures (including grazing and human presence) with short-term temporal regularity can lead to changes in animal activity rhythms. For instance, wildlife may reduce their activity during periods of high human presence (Nickel et al., 2020; Li et al., 2022c). In some cases, wildlife may avoid human-dominated habitats (e.g., farmlands and human settlements) altogether in response to long-term spatially consistent predation risk (Frid and Dill 2002, Tucker et al., 2018, Li et al., 2022c). These human activities pose major biodiversity conservation challenges in mountain. Moreover, environmental factors such as geographical differences, terrain complexity, and climatic variability (both spatially and temporally) influence the unique species communities of mountains (Rahbek et al., 2019) through complex effects on wildlife movement, foraging strategies, and survival (Jia et al., 2023; O’Kane and Macdonald, 2018). However, studies that focus on the spatiotemporal diversity patterns in response to human activity pressures remain scarce (Gaynor et al., 2018; Lwin et al., 2023; Nickel et al., 2020).

Wildlife habitat use exhibits seasonal variations, especially in mountainous landscapes with pronounced elevational ranges and seasonal climatic differences (Arias et al., 2022). In these landscapes, altitudinal migration is a common strategy for coping with seasonal environmental changes, allowing animals to utilize resources that vary across time and space (Bairlein, 2013; Wen et al., 2014). Altitudinal migration is a widespread phenomenon among mountain species, including resident birds, mammals, and, to a lesser extent, reptiles and insects (Hsiung et al., 2018), mainly influenced by external (e.g., resource availability, weather conditions, and predation risk) and internal (e.g., body size, competitive advantage, and the timing of arrival) factors (Hsiung et al., 2018). Previous research on altitudinal migration in large and medium-sized mammals has primarily focused on ungulate species (Ferrerías et al., 2016), with studies on carnivores relatively scarce and community-level research being almost nonexistent (Hsiung et al., 2018).

Beta diversity, which characterizes the spatial variation in species composition across different locations, is increasingly used in regional conservation planning (Angeler, 2013; Si et al., 2015). Spatial β diversity, representing community composition heterogeneity due to spatial gradients (Baselga, 2010), can be dissected into two primary components: turnover (differences in species composition between different locations) and nestedness (where species found at some locations are a subset of those found at other locations) (Baselga, 2010). Each component underscores different conservation approaches: turnover dominance implies that an area holds intrinsic conservation value, whereas nestedness dominance necessitates prioritizing regions with higher species richness (Si et al., 2015). Relatedly, temporal β diversity quantitatively measures community composition shifts over time, thereby serving as an indicator of seasonal biodiversity dynamics (Magurran et al., 2019; Shimadzu et al., 2015). In montane ecosystems, temporal β diversity is also influenced by human activities and elevation gradients (Elsen et al., 2017; Garcia-Navas et al., 2020). Investigating changes in mammalian communities across space and time is vital for identifying conservation priorities and formulating tailored conservation strategies.

Many human-altered ecosystems, including mountain ecosystems, which are important biodiversity hotspots, are increasingly losing their ecological integrity (Li et al., 2023; Pimm et al., 2014; Ripple et al., 2014; Tilker et al., 2020). The Gaoligong Mountains (GLGM), situated at

the intersection of three biodiversity hotspots (the Himalayas, Indo-Burma, and the Mountains of Southwest China) (Myers et al., 2000), harbor a rich community of terrestrial vertebrates, including 212 mammal (Li et al., 2024a), 796 bird (Wu et al., 2024), 107 reptile, and 84 amphibian (Wang et al., 2024) species, making it a key region for global biodiversity conservation efforts (Lan and Dunbar, 2009). The heterogeneity of the GLGM results from seasonal environmental changes and human activities (Li et al., 2022c; Zhou et al., 2023). Previous studies of animal diversity patterns in GLGM have been localized, focusing mostly on faunal species in the southern section (Gao et al., 2017; Li et al., 2022a; Liang et al., 2021) and small mammals at higher altitudes (Song et al., 2020). Research on large and medium-sized mammals has primarily focused on flagship species, such as the Gaoligong hoolock gibbon (*Hoolock tianxing*) (Fei et al., 2019; Fei et al., 2023), the Myanmar snub-nosed monkey (*Rhinopithecus strykeri*) (Wang et al., 2023; Yang et al., 2019), and the Himalayan takin (*Budorcas taxicolor*) (Li et al., 2020). Few studies have assessed the spatial and temporal distribution patterns of large and medium-sized mammals due to the poor terrain accessibility and cryptic habits of these species. Thus, the lack of systematic spatiotemporal distribution data, coupled with incomplete detection, can lead to an underestimation of species diversity (Si et al., 2018).

This study represents the first quantitative assessment of how human activities and environmental variables influence species distribution and community composition of large and medium-sized mammals in the GLGM, across different spatial and temporal dimensions. Our research was guided by two main objectives: (1) To investigate the extent to which human activities and environmental variables affect diversity patterns at different temporal scales in the GLGM; we hypothesized that the responses of species mean occupancy and community mean occupancy to human activities and environmental variables would differ across seasons because these variables vary significantly between seasons (Li et al., 2018; Quiroga-Pacheco et al., 2024). (2) To identify spatiotemporal patterns of β diversity in large and medium-sized mammals. Given the environmental heterogeneity due to the extensive span and pronounced climatic differences between the northern and southern regions of the GLGM, we hypothesized that spatially, the turnover components of β diversity would exceed the nestedness components (Baselga, 2010). Temporally, we hypothesized that lower elevations would gain more species in the dry season due to the harsh environmental conditions at higher elevations (Liang et al., 2021). By comprehensively analyzing the diversity patterns of large and medium-sized mammals in the GLGM and their determinants across spatial and temporal dimensions, we aimed to improve understanding of the ecological processes driving biodiversity and provide robust scientific support for targeted conservation strategies.

2. Material and methods

2.1. Study area and large and medium-sized mammals

Our study focused on large and medium-sized mammals weighing ≥ 0.5 kg (Gaynor et al., 2018; Li et al., 2022c) in the Gaoligong Mountains (GLGM), northwest Yunnan, China (24°56'N–28°30'N) (Fig. 1). The GLGM extends approximately 400 km from north to south (Li et al., 2000) and serves as a natural divide between the Dulong Jiang (Irrawaddy River) to the west and Nu Jiang (Salween River) to the east (Li et al., 2024a). The GLGM is higher at its northern end where the starting elevation and native forest on the eastern slope (Qiunatong) are 1560 m and 1800 m, respectively, and 1248 m and 1800 m, respectively, on the western slope (Qinlangdang). The southern end has lower elevations, with the starting elevation and native forest elevation on the eastern slope (Bawan) at 1040 m and 1500 m, respectively, and 920 m and 1500 m, respectively, on the western slope (Heikouping) (Zhou et al., 2023). The highest point, ‘Gawagapu Peak’, reaches 5128 m, resulting in a pronounced altitudinal gradient across 4208 m (Li et al., 2000; Wang

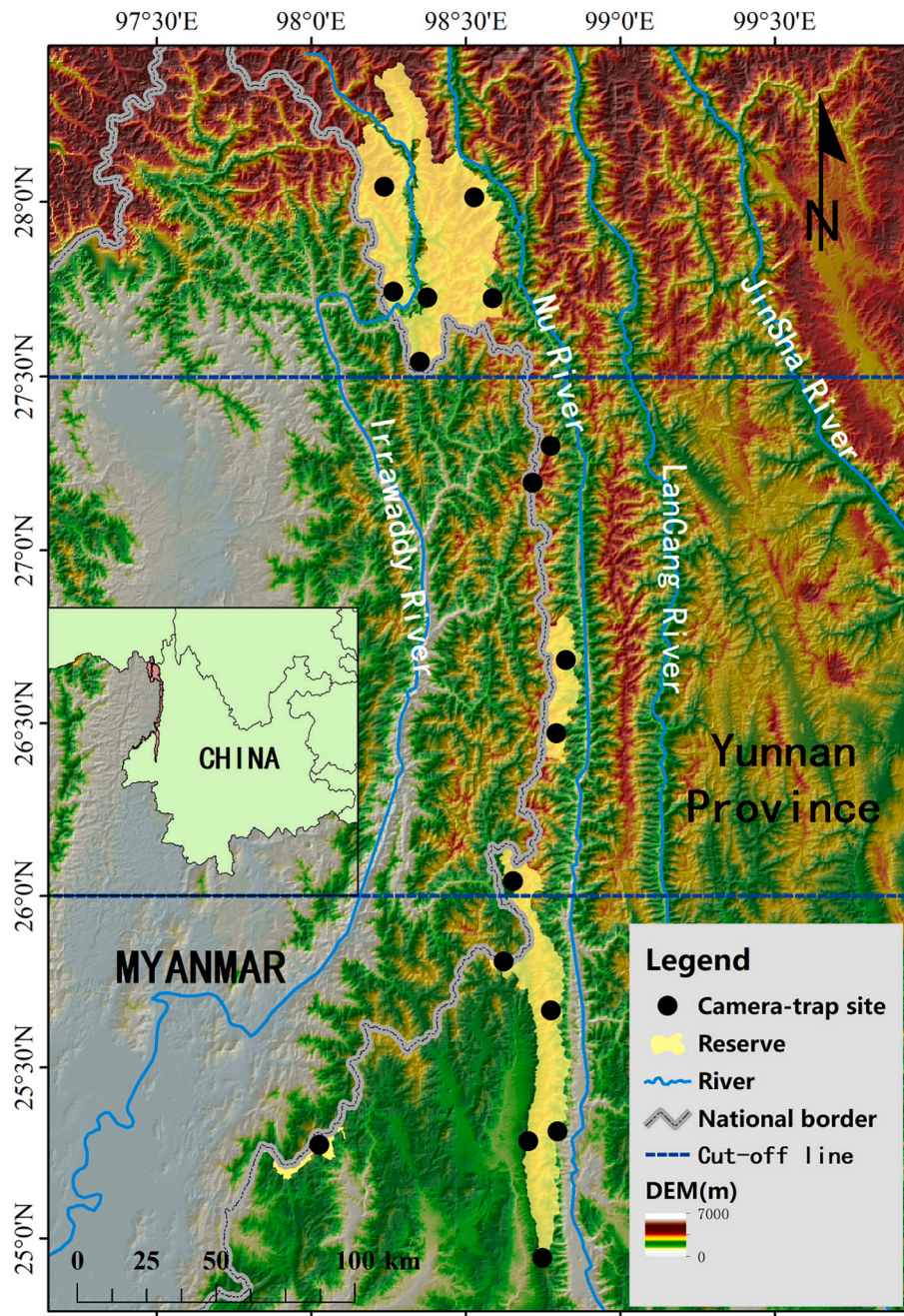


Fig. 1. A map of the study area (Gaoligong Mountains, northwest Yunnan, China) showing the 17 camera-trap sampling sites. The blue dashed line divides the study area into three sections: southern, middle, and northern. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

et al., 2007). Gaoligong Mountains span six climatic zones: southern subtropical, middle subtropical, northern subtropical, and alpine vertical zonal warm temperate, temperate and cold temperate zones (Wang et al., 2007; Zhou et al., 2023). The diverse climate and topography of the GLGM area supports a mosaic of habitats, ranging from Yunnan pine forests at lower elevations to a variety of ecosystems, including evergreen broadleaf forests, deciduous broadleaf forests, evergreen coniferous forests, bamboo forests, and meadows, at higher altitudes (Wang et al., 2007; Zhou et al., 2023). The hydrological cycle consists of two main seasons: the wettest months from July to September and the coldest and driest months from December to February. Human presence spreads across elevations during the wet season as grazing and foraging activities, such as collecting herbs and mushrooms, increase (Li et al., 2022c). In the dry season, harsh conditions (cool temperatures and snow

cover) restrict human activities to lower elevations (see Supporting Information Appendix Fig. S1). The landscape below 1600 m is predominantly characterized by human settlements and farmlands (Pan et al., 2019). A significant portion of the study area overlaps the Gaoligongshan National Nature Reserve, an area that benefits from systematic protection efforts, except below the lower boundary (2000 m, 2500 m in the middle section) (Xue et al., 1995; Zhou et al., 2023).

2.2. Data collection

From January 2019 to December 2022, we undertook a comprehensive systematic camera trapping survey targeting large and medium-sized mammals in the GLGM. Employing a stratified sampling strategy, the survey was structured with a sampling interval of 0.5° latitude, and

an enhanced density of sampling stations for comprehensive coverage. Based on variable protection levels and climatic conditions, we segmented the study area into three sections: the southern section (south of 26° N, designated as a reserve), the middle section (between 26° and 27.5° N, a predominantly unprotected area), and the northern section (north of 27.5° N, designated as a reserve) (Xue et al., 1995; Zhou et al., 2023). The survey included 17 distinct sampling sites, each considered an individual sampling unit. Given the high risk of camera trap theft in areas below 1600 m elevation (Liang et al., 2021; Pan et al., 2019), we deployed camera-traps from 1500 m to the ridge lines which still encompassed an elevational range of 2400 m. In line with our research objectives, we categorized altitudes into low (1500–2300 m), medium (2300–3100 m), and high (3100–3900 m) zones.

At each sampling site, we deployed 15 to 30 camera-traps (Loreda L510 series and Heron S3), ensuring a minimum of 500 m between camera-traps. We deployed 375 camera-traps across the 17 sampling sites. Camera-traps were placed between 0.5 and 1 m above the ground, mainly oriented in the direction of animal tracks and away from human pathways to minimize interference. Camera-trap sensitivity was calibrated to “low” to avoid false triggers. Upon activation, the camera-traps were configured to capture two photographs, with an interval between triggers set to 3 s. Species identification was conducted with reference to the classification and distribution of Chinese mammals (Andrew and Xie, 2009; Wei et al., 2022) to accurately assign taxonomic details to the recorded species. A camera day was defined as a consecutive 24-h period during which a camera trap operated at a given location, serving as a standardized measure for the duration and intensity of data collection. An independent event (either of wildlife or disturbance) was treated as all of the same kind of images taken by the same camera-trap within a day (Li et al., 2018).

2.3. Anthropogenic and environmental covariates

The spatiotemporal patterns of species distributions are significantly influenced by human activities and environmental variables. Previous research demonstrated that varying human activities exert distinct impacts on animal communities (Li et al., 2022b; Li et al., 2022c; Nickel et al., 2020). We employed a Human Modification Index derived from the Gradient of Human Modification Map (Kennedy et al., 2019) to characterize long-term, spatially consistent human impacts on faunal communities. The HM dataset is based on 13 anthropogenic impacts caused by five major human activities (agriculture, electrical infrastructure, energy production, human settlement, mining, and transportation) and thus provides a cumulative estimate of human modification of terrestrial ecosystems at a resolution of 1 km (Kennedy et al., 2019). This index is frequently utilized in analyzing the effects of human activities on regional wildlife diversity patterns (Boron et al., 2023; Gilbert et al., 2022; Li et al., 2024b). We also derived a Human Presence Index to quantify short-term human impacts, encompassing both direct (D: humans) and indirect (I: livestock, dogs) activities. This was calculated based on the frequency of independent occurrences of each activity type per unit time, expressed as: (D or I independent events/camera days) × 100.

We compiled a set of environmental covariates expected to impact terrestrial mammal diversity and distribution: (1) Normalized Difference Vegetation Index (NDVI) derived from Landsat 8; (2) terrain index (e.g., mean elevation and slope) derived from ASTGTM DEM v003; (3) precipitation obtained from the WorldClim database (<https://worldclim.org/>). Pearson's correlation coefficients were calculated for all pairs of covariates, and a threshold of $|r| < 0.8$ was used as a cutoff for dropping highly correlated variables (Table S1). Variables were standardized to ensure consistency of analysis (Dormann et al., 2013).

2.4. Occupancy model

A multi-region community model was employed to examine species

and community occupancy across camera-trap stations and sampling sites, and the corresponding correlation with human activities and environmental variables (Sutherland et al., 2016). A sampling unit time was recorded as the detection history of each species at every camera-trap station over a 24-h interval, with instances of species presence denoted as “1” and absence as “0”. Under the assumption that a species detection record (1 or 0) aligns with the Bernoulli stochastic process, we developed a hierarchical framework for the occupancy and detection parameters of species i within sampling site r and camera-trap station j :

$$w_{i,r} \sim \text{Bern}(\Omega_r)$$

$$Z_{i,r,j} \sim \text{Bern}(\Psi_{i,r,j} * w_{i,r})$$

$$y_{i,r,j,t} \sim \text{Bern}(p_{i,r,j,t} \times Z_{i,r,j})$$

The variable $w_{i,r}$ denotes the occurrence status of species i at sampling site r , and was assigned 1 for presence and 0 for absence. The variable Ω_r represents the mean occupancy of all species at sampling site r , reflecting the proportion of each species at sampling site r compared to the study area (Zipkin et al., 2010). $Z_{i,r,j}$ represents the presence state of species i at camera-trap station j at sampling site r , $\Psi_{i,r,j}$ represents occupancy of species i at camera-trap station j at sampling site r , $y_{i,r,j,t}$ represents the actual detection (1 or 0) of species i at sampling unit t of camera-trap station j at sampling site r , and $p_{i,r,j,t}$ corresponds to the detection probability (Dorazio et al., 2006).

The hierarchical occupancy model assumes that the population is closed and species occurrence data across camera-trap stations are independent of each other, which presents considerable challenges when applied to large and medium-sized mammals due to their high mobility. Consequently, we interpreted $\Psi_{i,r,j}$ as the probability of habitat utilization at camera-trap station j at sampling site r , as opposed to actual species occupancy, and $p_{i,r,j,t}$ as the intensity of habitat utilization rather than an accurate measure of species detection probability (Kays et al., 2020; MacKenzie et al., 2002). We used the same set of covariates at different scales to ensure comparability and consistency of the results, for a more holistic understanding of the impacts of human activities and environmental factors on diversity patterns.

At the sampling site scale, we incorporated the human modification index, human presence index, elevation, latitude, precipitation, slope, and NDVI as covariates. We then employed the Poisson link function with a logarithmic function to model the relationship between species richness (λ_r), environmental variables, and the human activity index across each survey sample:

$$\log(\lambda_r) = \beta_0 + \sum_{i=1}^n \beta_i \text{cov}_{ir}$$

At the scale of camera-trap station, we utilized human modification index, human presence index, elevation, latitude, precipitation, slope, and NDVI as covariates and modelled the relationship between species occupancy and detection and environmental variables and the human activities index at the species level. A logit function was applied to each camera-trap station (Li et al., 2022c):

$$\text{logit}(\Psi_{i,r,j}) = \gamma_0 + \sum_{i=1}^n \gamma_i \text{cov}_{j,r}$$

$$\text{logit}(p_{i,r,j,t}) = \delta_0 + \sum_{i=1}^n \delta_i \text{cov}_{j,r}$$

The posterior distribution, mean, and confidence intervals of the model parameters were derived using Bayesian inference and the calculation of species richness at the camera-trap station scale was based on the posterior distribution of the latent variables. The ‘jagsUI’ package (Kellner, 2024) was used for the correlation analysis in R 4.3.3 (Team, 2024).

2.5. Measuring space utilization

Variations in species composition across regions were assessed using spatial β diversity analysis (Palmeirim et al., 2018). We computed the Sørensen index of heterogeneity using the 'betapart' package (Baselga et al., 2023), with presence-absence data, thus enabling the analysis of β diversity (β SOR) along with its spatial nestedness (β SNE) and turnover (β SIM) components (Baselga, 2010). Furthermore, we computed temporal β diversity to assess shifts in community composition over time, drawing from repeated surveys across sites (Shimadzu et al., 2015). Species losses and gains across seasons were quantified based on species abundance data, and the Bray-Curtis dissimilarity index in the 'adespatial' package (Dray et al., 2023; Legendre, 2019). The statistical significance of the seasonal variations was assessed through 9999 random permutations for each elevation zone.

Species richness across elevation zones and seasons was quantified via the estimatedD function ($q = 0$) in the 'iNEXT' package (Hsieh et al., 2016). We also calculated sample coverage estimates to evaluate the comprehensiveness of our sampling effort for each altitude zone and season in 'iNEXT'. Moreover, we calculated species abundance-weighted mean elevation (AWME) over the two seasons:

$$AWME = \frac{\sum (N_j \times E_j)}{\sum N_j}$$

Where: N_j = abundance at elevation j and E_j = elevation at camera-trap station j .

The AWME served as an indicator of species mobility along the elevational gradient (Ocampo-Peñuela and Pimm, 2015). A difference in AWME between seasons indicates directional movement: upward (positive), downward (negative), or zero (no change) (Liang et al., 2021). For species recorded at more than five camera-trap stations, we also estimated the upper and lower elevation limits, elevation ranges, and AWME values between seasons, and tested their seasonal significance by t -test.

3. Results

3.1. Result at species level

We recorded 34 species of large and medium-sized mammals from 375 stations spanning 113,204 camera days of effort. The events exhibited considerable variation between species, ranging from a high of 2196 captures for the northern red muntjac (*Muntiacus vaginalis*) to a single record for the Myanmar snub-nosed monkey (Fig. 2, Table S2). A majority of these species (24) are designated as nationally protected animals under China's Wildlife Protection Law, while 13 are threatened according to the International Union for Conservation of Nature (IUCN) Red List criteria, including the Myanmar snub-nosed monkey as critically endangered, six as endangered, and six as vulnerable. The species occupancy varied significantly, from 0.61 for the Asiatic black bear (*Ursus thibetanus*) to a low of 0.01 for the forest musk deer (*Moschus berezovskii*). Only a few species, including the Himalayan takin (0.50), northern red muntjac (0.39), Malayan porcupine (*Hystrix brachyura*)

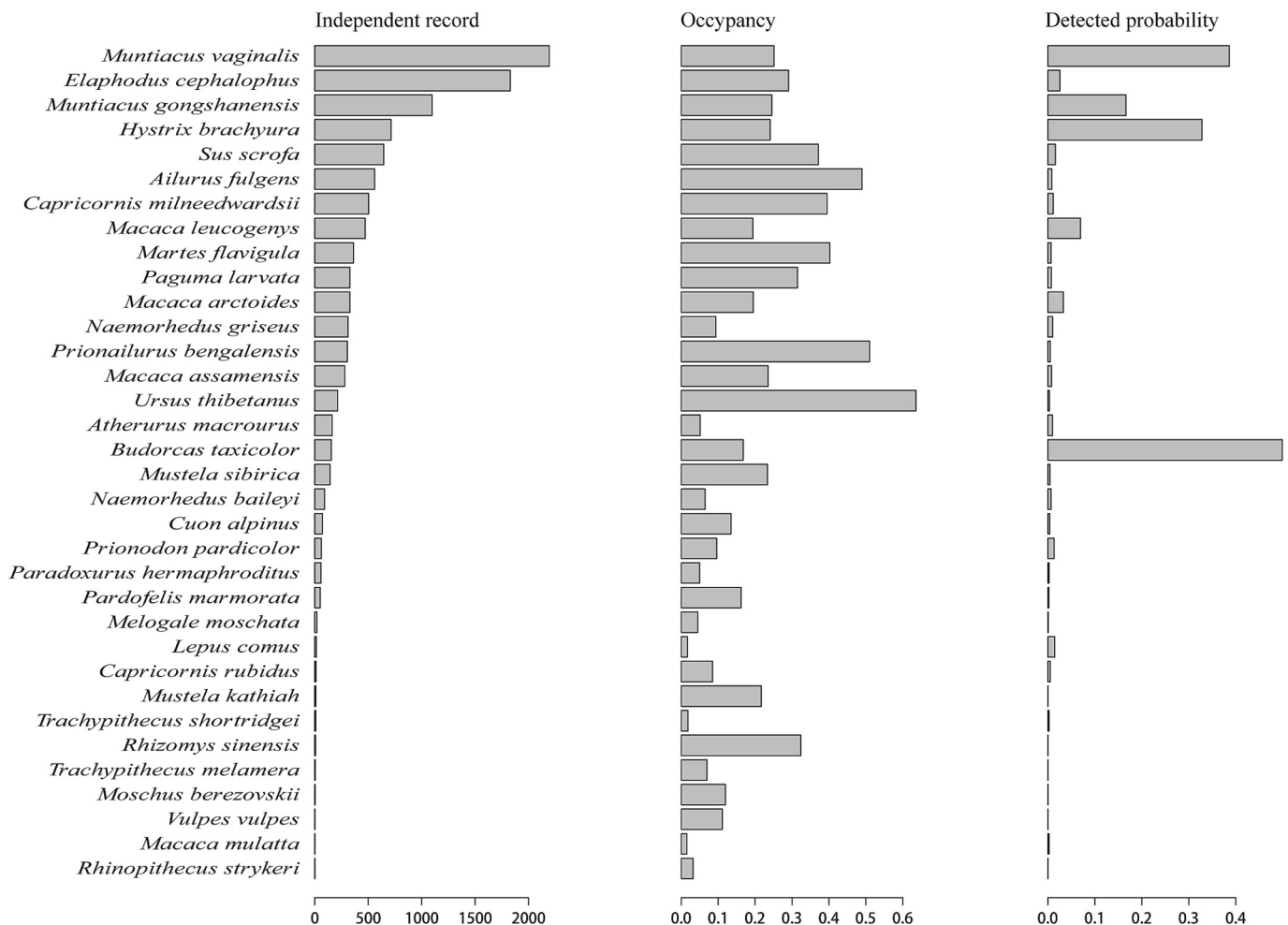


Fig. 2. Independent records, mean occupancy, and detection probabilities of large and medium-sized mammals based on camera trapping survey in Gaoligong Mountains.

(0.33), and Gongshan muntjac (*Muntiacus gongshanensis*) (0.17), exhibited high detection probabilities, with that of other species remaining below 0.1.

Distinct variations were observed in the impact of human activities index and environmental variables on species occupancy at camera-trap stations (indicative of habitat utilization probability) (Fig. 3). The mean occupancy of nine species were significantly affected by human modifications; seven had significant negative correlation with human modification and two showed a opposite response (Table S3). The mean occupancy of four species, including three primate species and the yellow-throated marten (*Martes flavigula*), was significantly negatively influenced by human presence. Among the environmental variables, latitude significantly influenced the mean occupancy of 16 species, with 13 species negatively impacted, and three species positively impacted. Elevation had significant effects on the mean occupancy of 10 species. Among these, seven species showed a substantial negative correlation, whereas three species demonstrated a positive association. Annual precipitation had a significant effect on the mean occupancy of nine species; negative effect on two species and positive effect on seven species. NDVI had a positive influence on the mean occupancy of the Chinese serow (Table S3).

3.2. Diversity patterns across spatial and temporal dimensions

The correlation between latitude and mean occupancy was statistically significant throughout the year ($R = -0.77$, $df = 364$, $p < 0.001$) (Fig. 4a). Elevation was also significantly negatively associated with the mean occupancy of mammals ($R = -0.34$, $df = 364$, $p < 0.001$). Annual precipitation ($R = 0.68$, $df = 364$, $p < 0.001$), NDVI ($R = 0.42$, $df = 364$, $p < 0.001$) and the human modification index ($R = 0.37$, $df = 364$, $p < 0.001$) were significantly positively correlated with the mean occupancy of mammals (Fig. 4a). The response patterns of mean occupancy to the human activity indices and environmental variables in the dry season and wet season were similar to those of the whole year. Specifically, the effect of elevation on the mean occupancy of mammals in the dry season

($R = -0.59$, $df = 290$, $p < 0.001$) was greater than that in the wet season ($R = -0.32$, $df = 274$, $p < 0.001$) (Fig. 4b, c).

At the community level (sampling site scale) considered across the entire year, there was a significant negative correlation between the community mean occupancy and latitude (mean = -0.4 ± 0.20 , 95 % confidence interval: -0.80 to -0.02), indicating that with increasing latitude, the community mean occupancy decreased, and the probability of habitat utilization at camera-trap stations was reduced (Table 1). Latitude had the same effect (mean = -0.37 ± 0.19 , 95 % confidence interval: -0.75 - 0.00) during the dry season. During the wet season, latitude also had a negative effect on the community mean occupancy (mean = -0.42 ± 0.26 , 95 % confidence interval: -0.92 - 0.09), although the effect size was not statistically significant and the parameter posterior distribution indicated a negative estimated coefficient. During the dry season, elevation (mean = -0.17 ± 0.12 , 95 % confidence interval: -0.40 - 0.08), the human modification index (mean = -0.17 ± 0.13 , 95 % confidence interval: -0.44 - 0.09) and the human presence index (mean = -0.09 ± 0.05 , 95 % confidence interval: -0.21 - 0.02) all had negative effects on community mean occupancy (Table 1).

3.3. Space utilization of large and medium-sized mammals

Compared 800 m elevation bands, species composition at high (3100–3900 m, $p = 0.474$) and middle (2300–3100 m, $p = 0.680$) elevations did not significantly vary from wet season to dry season, whereas the variation in species composition at low elevations (1500–2300 m, $p = 0.052$) was close to significant (Fig. 5a). Comparing 200 m bands, species gain from wet to dry season dominated temporal β diversity at 1700–2300 m elevations, and species loss dominated temporal β diversity at 2300–3900 m (Fig. 5b and Table S4). This implies that the middle and high-altitude species move to low altitudes during the dry seasons. Seasonal variations in species composition were predominantly driven by species loss, rather than species gain, especially at the highest and lowest altitudes. Across altitudinal zones, significant changes in community composition were observed in the highest zone (3300–3900

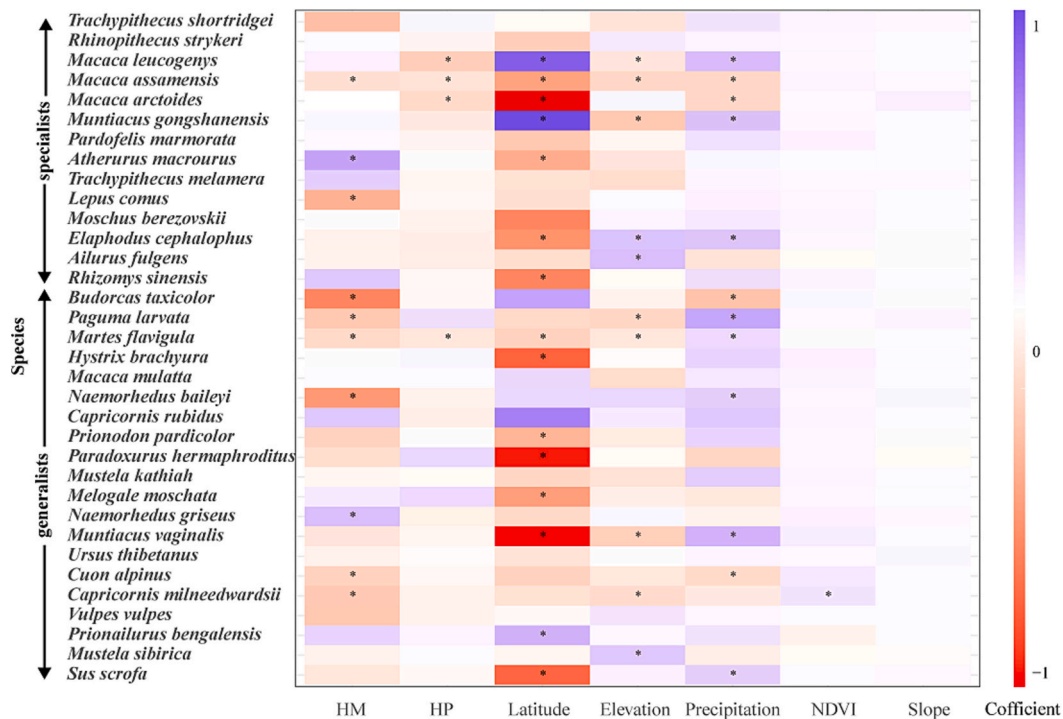


Fig. 3. Occupancy response of large and medium-sized mammals to human activities and natural environmental variables in Gaoligong Mountains. ‘*’ means significant. Confidence intervals representing 97.5 % are all greater than or all less than zero (HM: human modification; HP: human presence; NDVI: Normalized Difference Vegetation Index).

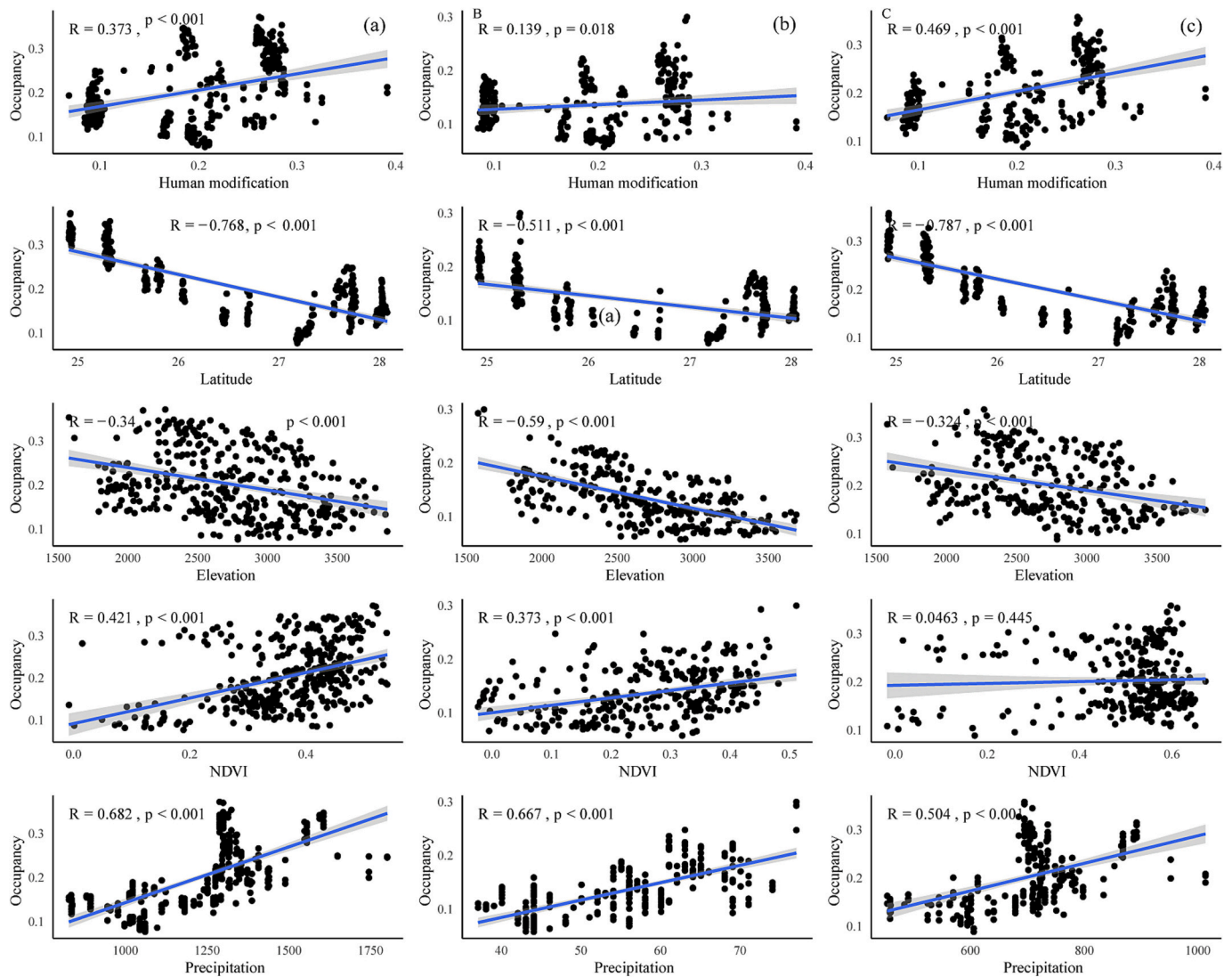


Fig. 4. Patterns of large and medium-sized mammal diversity responses to human activities and natural environmental variables in Gaoligong Mountains across seasons. Annual (a), dry season (b), wet season (c) (NDVI: Normalized Difference Vegetation Index).

Table 1

Response of community mean occupancy to human activities and environmental variables across seasons.

	Annual	Dry	Wet
HM	-0.14, (-0.36–0.10)	-0.17, (-0.44–0.09)	-0.06, (-0.30–0.18)
HP	-0.07, (-0.19–0.07)	-0.09, (-0.21–0.02)	-0.05, (-0.25–0.11)
Latitude	-0.4, (-0.80–0.02)	-0.3, (-0.75–0.00)	-0.42, (-0.92–0.09)
Elevation	-0.0, (-0.21–0.10)	-0.17, (-0.40–0.08)	-0.07, (-0.26–0.13)
Slope	0.03, (-0.03–0.10)	-0.02, (-0.13–0.08)	0.10, (0.01–0.19)
NDVI	0.07, (-0.01–0.16)	0.02, (-0.07–0.12)	0.05, (-0.09–0.18)
Precipitation	0.11, (-0.06–0.31)	0.06, (-0.16–0.31)	0.07, (-0.14–0.26)

Note: HM: Human modification; HP: Human presence; NDVI: Normalized Difference Vegetation Index. Responses are shown as effect means with 95 % confidence intervals (upper, lower). We consider a response significant if the 95 % confidence interval does not overlap zero.

m; $p = 0.029$, $p = 0.004$, $p < 0.001$) and the lowest (1500–1700 m; $p < 0.001$) zones (Fig. 5b and Table S4).

Twenty species were recorded at more than five camera-trap stations per season (Table S5). At the species level, there was a notable decrease in AWME for most species during the dry seasons, compared to the wet seasons. Among them, the AWME of the Himalayan takin, spotted linsang (*Prionodon pardicolor*), marbled cat (*Pardofelis marmorata*) and Chinese serow decreased by ≥ 300 m. A few species, including the Gongshan muntjac, exhibited minimal changes in their elevation. The

Siberian weasel (*Mustela sibirica*), masked palm civet (*Paguma larvata*), and tufted deer were the only species to exhibit an upslope movement during the dry seasons (Fig. 6). However, at the community level, the upper elevation limit ($p < 0.001$) and the elevation range ($p = 0.011$) of the dry season community showed significant downslope shifts compared with the wet seasons, unlike the lower elevation limit ($p = 0.542$) and the AWME ($p = 0.062$).

At the scale of the entire GLGM, the spatial turnover component ($\beta\text{SIM} = 0.68$) predominated over the nestedness component ($\beta\text{SNE} =$

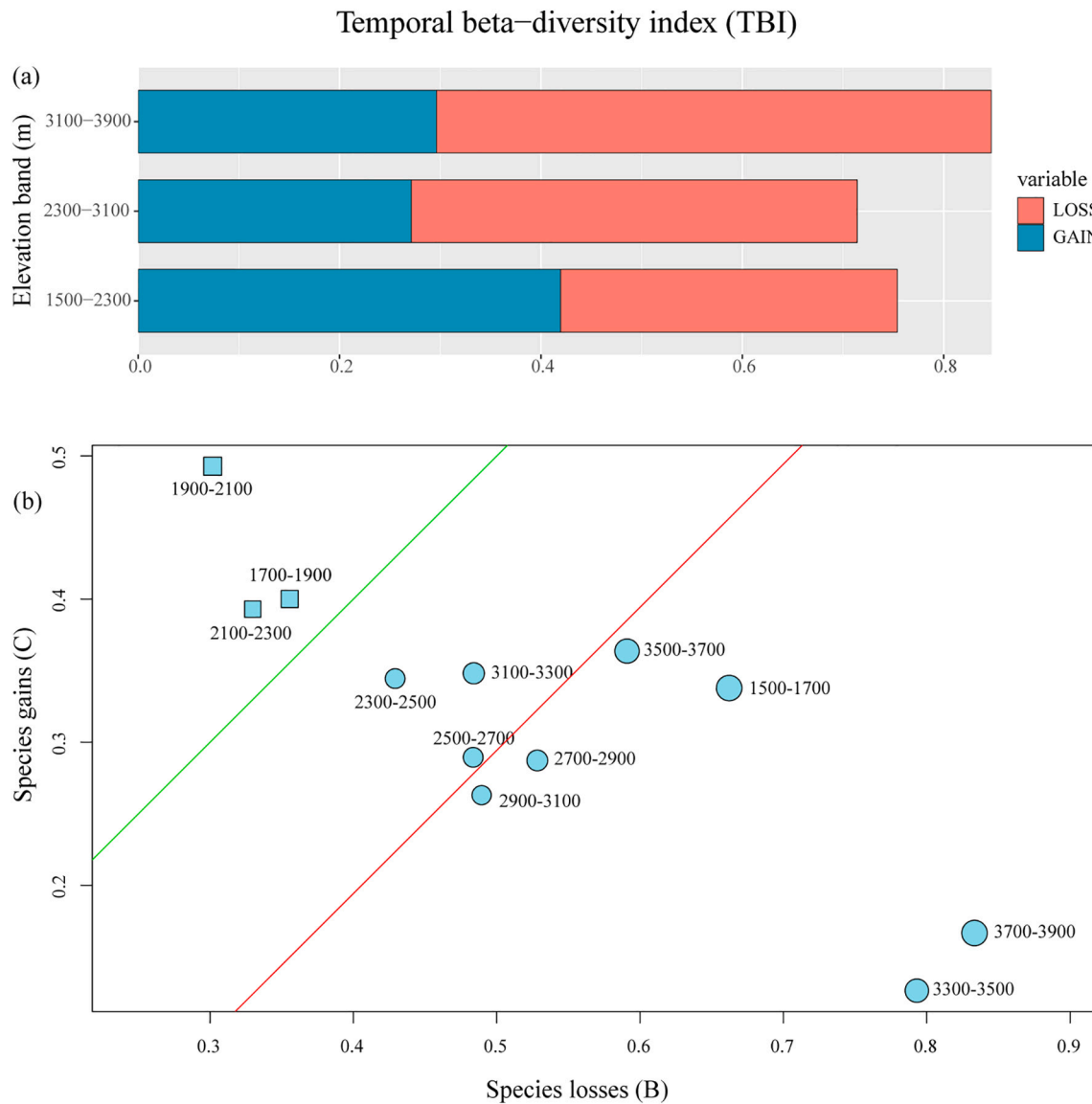


Fig. 5. Decomposition of temporal β -diversity into species gain and loss: a) Species loss(B) and species gain(C) at all elevations (800 m bands) from wet seasons to dry seasons; b) B–C plots of species abundance data for 12 altitude zones (200 m bands). The green line indicates the balance between species gains and losses, and parallel red lines intersect the centers of mass of all altitude zones. A location below the green line indicates that species losses outweigh species gains, on average, from wet seasons to dry seasons. The size of the symbols indicates the relative value of the temporal β diversity index. Circles ($n = 9$) represents the altitude zone dominated by species loss, and squares ($n = 3$) represents the altitude zone dominated by species gain. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

0.11), and the Sørensen heterogeneity index was 0.79. At the local scale, the turnover component of spatial β diversity was greater than the nestedness component in the northern ($\beta\text{SIM} = 0.34$, $\beta\text{SNE} = 0.11$, Sørensen = 0.45), middle ($\beta\text{SIM} = 0.44$, $\beta\text{SNE} = 0.18$, Sørensen = 0.62), and southern ($\beta\text{SIM} = 0.24$, $\beta\text{SNE} = 0.12$, Sørensen = 0.36) sections. Across the entire GLGM, spatial β diversity exhibited the most pronounced change, while at the local scale, the middle section of the GLGM portrayed the most pronounced change.

4. Discussion

Despite vast and accumulating evidence of widespread human impacts on wildlife diversity and distributions, our understanding of how human disturbance reshapes species distribution along elevational gradients across different seasons remains limited. By integrating systematic camera trap data with hierarchical occupancy models, we were able to disentangle the effects of human activities on wildlife distributions at

species and community levels. Our results suggest that lower altitude zones are crucial for wildlife species particularly during dry seasons.

4.1. Species and community response patterns

Inconsistent with our initial hypothesis, the mean occupancy of large and medium-sized mammals in GLGM responded uniformly to human activities and environmental variables at the species level throughout the year, during the dry season, and during the wet season, but different variables varied in degree of influence across different seasons (Fig. 4a, b, c). Species' mean occupancy is an index of the probability of habitat utilization around the camera trap. Since the area covered by each camera trap was small, the variation in environmental variables between seasons was quantitative, and species' habitat use preferences did not change significantly (Martin et al., 2017). In contrast, and in agreement with our predictions, the response pattern of the community mean occupancy to human activities and environmental variables was

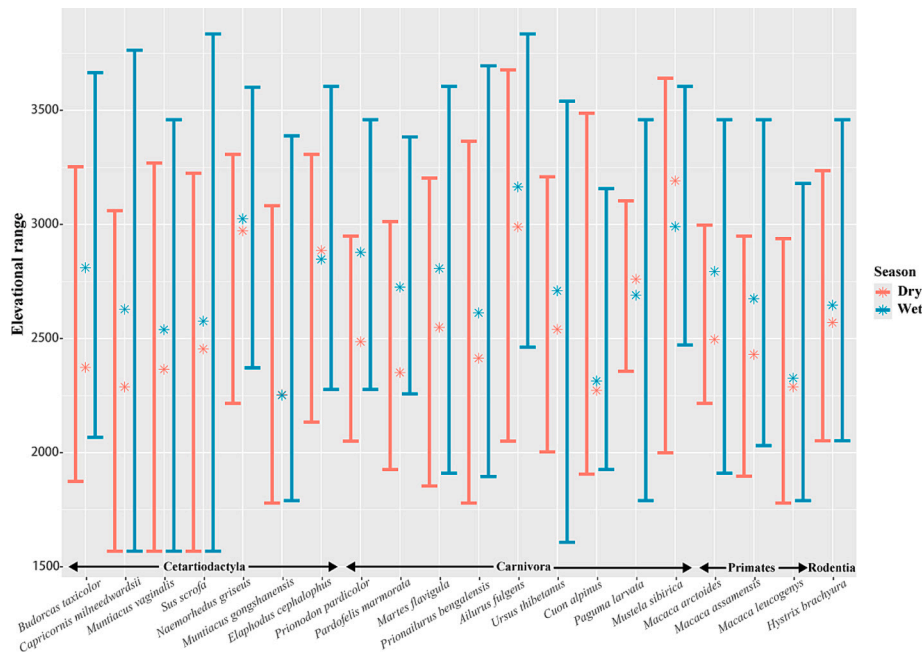


Fig. 6. Elevation ranges of large and medium-sized mammals in the dry and wet seasons. Note: '*' represent species abundance-weighted mean elevation (AWME) over the dry or wet season.

inconsistent at the community level. Although latitude had a similar negative effect on community mean occupancy across seasons, community mean occupancy was also negatively affected by elevation, human modification and human presence in the dry season. This means that the effects of seasonal changes cannot be ignored at the community level (Jackson and Fahrig, 2015; Quiroga-Pacheco et al., 2024).

We observed an increasing trend in the probability of utilization of space by large and medium-sized mammals along human modification gradients, consistent with previous findings in mountain forests of southwest China (Li et al., 2022c). Most study areas were located in protected zones, where human activities were at low to medium levels due to conservation management interventions. This suggests that the region's relatively low level of human modification increases habitat patch heterogeneity, potentially increases the probability of species habitat utilization (Catford et al., 2012). However, further increase in human activities would reduce species occupancy (Polaina et al., 2018). Correlation analysis of large and medium-sized mammal richness with the human modification index in Southwest China have shown a positive relationship, supporting the moderate disturbance hypothesis (Li et al., 2022c). However, we found that not all species exhibited a positive response to human activities, and the interaction between species habitat use and the level of human activities was mixed, showing both increases and decreases, suggesting that human activities enhance the relative abundance of certain adaptable species while diminishing that of more sensitive ones (Nickel et al., 2020; Tucker et al., 2021; Wisz et al., 2013).

The mean occupancy increased with annual precipitation and NDVI, supporting the energy and predation risk hypotheses (Suselbeek et al., 2014), which suggest that higher precipitation and NDVI provide more energy and cover, reducing predation and human disturbance. However, during the wet season, increased NDVI did not significantly affect species occupancy, suggesting that food resources were less limiting. Elevation also had a strong negative effect on occupancy, particularly during the dry season, as lower temperatures and snowfall at higher altitudes drove species to lower elevations (Garcia-Navas et al., 2020). Additionally, mean occupancy decreased with increasing latitude, presumably due to extreme cold, which serves as a significant environmental filter for species (Brodie, 2019; Zhou et al., 2023), though this effect weakened in the dry season as temperatures homogenized across

the mountain range.

4.2. Space utilization and human impact

Seasonal variations in the community of large and medium-sized mammals in the high-altitude zones (3300–3900 m) aligned with our predictions, revealing marked compositional changes between the dry and wet seasons. However, we observed significant changes in the community composition at GLGM's low-altitude zones in the dry seasons. This indicates that despite substantial human activities during the dry seasons, large and medium-sized mammals in these areas still descend to lower altitudes. In the absence of intense human activities, it is plausible that these species could migrate even further downslope. Habitats at lower elevations of GLGM play a pivotal role in species conservation, and their significance may have been previously underestimated (Liang et al., 2021).

During the dry seasons, with the exception of tufted deer, Siberian weasel and masked palm civet we observed a marked downward shift in elevations utilized by most species as measured by AWME. Temporal β -diversity analysis also revealed an increase in species richness in the lower elevation range (1700 to 2300 m) during the dry season, with a peak in species richness observed between 2100 and 2300 m (Fig. S2). The primary factors driving this altitudinal migration are likely harsh climatic conditions and the scarcity of food resources (Hsiung et al., 2018; Liang et al., 2021). The lower boundary of the GLGM protected area typically exceeds 2000 m, rising to over 2500 m in the middle section (Zhou et al., 2023). This elevation suggests that the boundary does not adequately accommodate the spatial requirements of large and medium-sized mammals for altitudinal migration and secure wintering grounds. We recommend that the lower boundary of the under-construction GLGM National Park be reduced to enhance the conservation of large and medium-sized mammals. Given the needs of local communities and indigenous livelihoods, additional research is needed to identify representative areas that could be made available for prioritized protection.

Consistent with our hypothesis, the spatial β diversity turnover was substantially greater than the nestedness, indicating that localized regions of the GLGM hold significant conservation values (Baselga, 2010; Patterson and Atmar, 1986; Si et al., 2015). In particular, the middle

section of the GLGM, which is mostly unprotected, presented the highest β diversity in our analyses. Spatially, the areas that comprise the protection gap of the Gaoligongshan Reserve warrant increased attention because of their ecological significance. To facilitate habitat connectivity for key species of conservation concern, the currently unprotected areas of GLGM should be considered for protection in the forthcoming establishment of GLGM National Park.

4.3. Conservation strategies

Based on temporal patterns, the conservation strategy for large and medium-sized mammals in the GLGM requires particular focus during the dry seasons (December, January, and February). The temporal patterns also highlight the importance of low-altitude zones for mammal conservation. The study revealed that in winter, there is a notable contraction in the elevational distribution of large and medium-sized mammals, with a tendency to cluster in mid and low-elevation zones near human settlements and agricultural lands. Depending on the local sociocultural and economic dynamics, a refined management approach can be adopted particularly in winter. Emphasis should be on strengthening control over human presence in winter, a period of heightened survival pressure for animals, and efforts should be made to minimize human, livestock, and dog activities. This approach can help alleviate the survival pressure on large and medium-sized mammals and aid in the protection of rare and endangered species during challenging periods.

In the GLGM, a latitudinal gradient involves not only changes in climate, geographical location and other environmental factors but also changes in human activities, such as the population, economic development level, and ethnic composition (Wang et al., 2007; Zhou et al., 2023). Latitude is an intuitive representation of these highly heterogeneous, more complex predictors of the mammal community ecology dynamics (Fig. S3). Although the latitude range is only 4°, it underlies a high turnover in human activities and environmental factors that combine to make the GLGM a highly ecologically heterogeneous mountain range. Previous studies have treated highly heterogeneous mountain ranges such as the GLGM as single ecological units with uniform conservation strategies (Liang et al., 2021; Pan et al., 2019). Further research could identify suitable locations to separate these heterogeneous areas and propose specific and targeted conservation strategies to achieve more targeted and efficient biodiversity protection.

5. Conclusion

This study delineated the key factors influencing the diversity patterns of large and medium-sized mammals in the GLGM, revealing varied responses of these mammalian species to human activities and environmental factors across seasons. Our findings demonstrate the importance of integrating spatial heterogeneity and the temporal dimension into conservation strategies, highlighting the pivotal role of low-altitude habitats in sustaining wildlife. The assessment underscores the necessity for comprehensive, large-scale investigations to refine the understanding of these dynamics. The study also highlights the need for effective conservation initiatives that focus not only on spatial specificity but also on periods of increased vulnerability for sensitive species in areas with significant seasonal environmental variation. The findings suggest that conservation managers should establish different prioritized conservation assessment objectives with spatial and temporal specificity, accompanied by specific policies and management strategies. While the findings are specific to the GLGM range, they have broader conservation relevancies to other low-altitude, densely populated mountainous ecosystems (Li et al., 2018).

CRedit authorship contribution statement

Wenqiang Hu: Writing – review & editing, Writing – original draft,

Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xueyou Li:** Writing – review & editing, Visualization, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Kenneth Otieno Onditi:** Writing – review & editing, Validation. **Hongjiao Wang:** Writing – review & editing, Resources, Data curation. **Wenyu Song:** Writing – review & editing, Resources. **Zhechang Hu:** Writing – review & editing, Validation, Software. **Changzhe Pu:** Writing – review & editing, Resources, Data curation. **Mingjing Pu:** Writing – review & editing, Resources, Data curation. **Yun Xiong:** Resources, Data curation. **Suhua Yang:** Resources, Data curation. **Lin Zhu:** Resources, Data curation. **Jianqiang Yin:** Resources, Data curation. **Lifan He:** Resources, Data curation. **Haoyang Song:** Resources, Data curation. **Yuetai Li:** Resources, Data curation. **Ge Gao:** Resources, Data curation. **William V. Bleisch:** Writing – review & editing. **Xuelong Jiang:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The study was supported by the National Key R&D Program of China (2022YFC2602500, 2022YFC2601200), the National Natural Science Foundation of China (32370562), the Major Science and Technique Programs in Yunnan Province (202102AA310055), the Science and Technology Basic Resources Investigation Program of China (2021FY100200), the West Light Foundation of the Chinese Academy of Sciences, the Yunnan Provincial Youth Talent Support Program (YNWR-QNBJ-2020-127), and the China Biodiversity Monitoring and Research Network (Sino Bon). We thank the Gaoligongshan National Nature Reserve Administration, the Tongbiguan Nature Reserve Administration, and the Forestry and Grassland Office of Nujiang Lisu Autonomous Prefecture and Baoshan City for their project approval and logistical support. We thank Yuejiang Yu, Dazhou Peng and other 52 guides for their help in the field work.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biocon.2024.110863>.

Data availability

The data that support the findings of this study are openly available in Data Dryad (<https://datadryad.org/stash/share/0Q9D6175eun3oE7uOgJkc6YVEmlkRPy-YXAm518tQI>).

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